AI AGENTS IN FINANCE: A SURVEY

by

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ABSTRACT

The rapid advancement of large language models (LLMs) and autonomous agent frameworks has catalyzed a new paradigm in computational finance. AI agents, which are systems capable of perceiving financial environments, reasoning over complex data, and executing actions, are being deployed across trading, portfolio management, risk assessment, fraud detection, and advisory services. This report has two parts. The first is a survey of AI agents in finance, covering both reinforcement learning-based agents and LLM-powered autonomous agents. We propose a taxonomy along two dimensions: agent design (single-agent, multi-agent, memory-augmented, tool-augmented, and reasoning-and-reflection architectures) and financial application domain (trading, portfolio management, risk and fraud, advisory, and compliance). The survey identifies a gap: existing LLM-based financial agents operate primarily at the individual asset level without explicit macroeconomic transmission-mechanism reasoning, while macro-focused LLM systems are not directly connected to calibrated cross-asset allocation signals.

The second part addresses this gap. We propose **Macro-MOSAIC** (Mixture-Of-Schools Analogy-Informed Chains), a four-stage pipeline designed to bridge qualitative macro event interpretation and quantitative asset allocation: (1) dual-channel retrieval over a historical macro episode database, (2) multi-school causal chain generation via school-specific fine-tuned LLMs (Keynesian, Austrian, Monetarist), (3) Bayesian aggregation with regime-conditional weighting and Platt scaling, and (4) fractional Kelly asset allocation. We report preliminary results on two *completed*

components—Stages 1 and 2—while Stages 3 and 4 remain planned work. First, we construct a structured database of 141 macro episodes (1971–2025) with 35 FRED indicators and 22 tracked instruments per episode, yielding 13,571 asset reaction data points. Second, we fine-tune school-specific LoRA adapters on Qwen2.5-7B-Instruct and evaluate on 96 held-out events. Fine-tuning improves school-style vocabulary consistency by 59.3% on average (Austrian +118.8%, Monetarist +129.7%), reduces cross-school contamination by up to 100%, and increases inter-school signal diversity (Jaccard distance 0.853 vs. 0.778 for base); these results are best interpreted as input-quality validation for Stage 3, not as evidence of end-to-end causal reasoning or trading performance. We conclude with a detailed plan for the remaining stages, ablation experiments, and end-to-end evaluation on both reasoning benchmarks (EconCausal, EconLogicQA) and out-of-sample backtests (2023–2025).

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Chapter 1

Introduction

The intersection of AI and finance has a long history, from early expert systems for credit scoring to statistical arbitrage models powered by machine learning. The emergence of large language models (LLMs) and autonomous agent frameworks has introduced a new capability: AI systems that perceive, reason, plan, and act within complex financial environments with increasing autonomy.

An AI agent in finance can be defined as a computational system that observes financial data (prices, news, reports, sentiment), maintains internal state or memory, reasons about market conditions, and takes actions such as placing trades, generating reports, or flagging risks. This covers a wide spectrum, from deep reinforcement learning (DRL) agents trained to optimize portfolio returns [Liu et al., 2022b,a] to LLM-powered multi-agent frameworks that simulate the collaborative dynamics of professional trading firms [Xiao et al., 2024].

Part I: Literature Review

Several recent surveys have examined LLM-based financial agents from different angles: Ding et al. [2024] focus specifically on trading applications, Nie et al. [2024] provide a broad overview of LLMs in financial applications, and Dong et al. [2025]

bridge research and real-world deployment. Building on these, the first part of this report reviews the literature with a specific orientation toward the design of our proposed system: we organize the space jointly along *agent architecture* (single-agent, multi-agent, memory-augmented, tool-augmented, reasoning-and-reflection) and *financial application domain* (trading, portfolio management, risk and fraud, advisory, compliance), so that the design choices made in Macro-MOSAIC can be explicitly situated against existing patterns. The review is organized around the following questions:

- **RQ1:** What data sources and modalities are used by AI agents in finance?
- **RQ2:** What agent design patterns have been proposed, and how do they differ architecturally?
- **RQ3:** How are these designs applied across different financial domains?
- **RQ4:** How are financial AI agents evaluated, and what benchmarks exist?
- **RQ5:** What are the key open challenges and promising future directions?

Part II: Gap, Proposed System, and Preliminary Results

The survey surfaces a specific gap at the boundary between equity-level agents and macroeconomic reasoning. Existing systems such as FinGPT [Yang et al., 2023], TradingAgents [Xiao et al., 2024], and FINCON [Yu et al., 2024] generate asset-level signals from news, indicators, and fundamentals, but operate primarily at the individual asset level without explicit macroeconomic transmission-mechanism reasoning. Conversely, macro-focused LLM systems such as EconAgent [Li et al., 2024] simulate or forecast macro dynamics but are not directly connected to calibrated cross-asset allocation signals; equity analysis systems such as MarketSenseAI [Fatouros et al.,

2025] similarly lack explicit macro transmission-mechanism reasoning. Three gaps motivate this work: (i) no system we are aware of provides *explicit, auditable multi-school parallel reasoning chains* in which distinct economic frameworks independently analyze the same macro event and are formally aggregated; (ii) *historical analogy retrieval at the macro-episode level* is absent from the systems we surveyed; (iii) no system connects qualitative macro interpretation to calibrated probability distributions and cross-asset signals in a *verified end-to-end pipeline* evaluated against both reasoning benchmarks and realized outcomes.

To address these gaps, we propose **Macro-MOSAIC** (Mixture-Of-Schools Analogy-Informed Chains), a four-stage pipeline:

1. **Historical Analogy Retrieval.** Dual-channel retrieval over 141 macro episodes (1971–2025), combining semantic similarity over event narratives with Mahalanobis distance over 35-dimensional FRED macro state vectors.

2. **Multi-School Causal Chain Generation.** Three school-specific LoRA adapters (Keynesian, Austrian, Monetarist) independently interpret the retrieved analogies and produce structured causal chains mapping transmission mechanisms to asset impacts.

3. **Bayesian Aggregation and Calibration.** Regime-conditional weighting based on rolling Brier Scores, followed by Platt scaling trained on resolved Polymarket and Metaculus questions.

4. **Asset Allocation Signal Generation.** Fractional Kelly allocation over 8 liquid instruments, driven by scenario-conditional return forecasts.

We report preliminary results on two completed components. The *episode database* (Section 4.1) contains 141 episodes across 8 categories and 19 transmission channels, with 35 FRED indicators and 22 instruments per episode, producing 13,571 asset reaction data points; 117 episodes additionally have LLM-generated narrative sum-

maries and three parallel school-specific interpretations (Keynesian, Austrian, Monetarist). The *school-specific adapter evaluation* (Section 4.2) fine-tunes LoRA adapters on Qwen2.5-7B-Instruct using 500 synthetic agent trajectories per school, and evaluates on 96 held-out events. Fine-tuning yields a 59.3% average gain in school-style vocabulary consistency (Austrian +118.8%, Monetarist +129.7%), reduces cross-school contamination by 51–100%, raises structured output block presence from 51–81% to 93–99%, and produces a higher inter-school Jaccard diversity (0.853 vs. 0.778); these results constitute input-quality validation for Stage 3 rather than evidence of end-to-end system performance.

Organization

The remainder of this report is organized as follows. Chapter 2 presents the literature review, covering background (§2.1), data sources and modalities (§2.2), agent design patterns (§2.3), applications (§2.4), and benchmarks (§2.5). Chapter 3 presents the research methodology: gap analysis (§3.1), problem formulation (§3.2), and the Macro-MOSAIC system architecture (§3.3). Chapter 4 reports preliminary results on the episode database and school-specific adapter evaluation. Chapter 5 details the plan and schedule for the remaining stages, ablation experiments, and end-to-end evaluation.

Chapter 2

Literature Review

2.1 Background

2.1.1 Foundations of AI Agents

An AI agent is typically defined as a system that perceives its environment, maintains internal state, and takes actions to achieve specified goals [Wooldridge and Jennings, 1995, Russell and Norvig, 2010]. In the context of LLM-based agents, recent frameworks have formalized the agent loop as a cycle of observation, reasoning (often via chain-of-thought prompting), action selection (potentially using external tools), and memory update [Wang et al., 2024a, Xi et al., 2025].

Key architectural components of modern AI agents include: (1) *Perception* modules that process environmental inputs, including text, numerical data, and visual information; (2) *Memory* mechanisms encompassing short-term (context window) and long-term (retrieval-augmented) storage for maintaining state across interactions [Park et al., 2023]; (3) *Reasoning* capabilities such as chain-of-thought [Wei et al., 2022], tree-of-thought [Yao et al., 2023], or reflection-based deliberation; (4) *Action* interfaces including tool use, API calls, and code generation [Schick et al., 2023]; and (5) *Planning* modules that decompose complex goals into subgoals [Huang et al.,

2024].

2.1.2 From Quantitative Models to Agents in Finance

The evolution of AI in finance can be traced through several paradigms. Early quantitative finance relied on statistical models such as ARIMA and GARCH for time series prediction. The rise of machine learning brought random forests, gradient boosting, and neural networks for tasks like credit scoring, fraud detection, and price prediction [Dixon et al., 2020].

Deep reinforcement learning (DRL) represented a significant shift toward agent-based financial systems. DRL agents learn trading policies by interacting with simulated market environments, optimizing objectives such as cumulative returns or Sharpe ratio [Hambly et al., 2023]. The FinRL framework [Liu et al., 2022b,a] established the first open-source library for financial DRL, supporting algorithms such as PPO, A2C, DDPG, SAC, and TD3 across stock trading and portfolio management tasks. FinRL-Meta [Liu et al., 2022a] further extended this with near-real market environments and comprehensive benchmarking.

The most recent paradigm shift involves LLM-powered agents that leverage natural language understanding and reasoning capabilities. These agents can process unstructured financial text, reason about market conditions in natural language, and integrate multiple data modalities [Yang et al., 2023, Zhang et al., 2024]. This transition from numerical-only DRL agents to language-capable autonomous agents constitutes the central focus of this survey.

2.1.3 Multi-Agent Systems

Multi-agent systems (MAS) have a long history in computational finance, from agent-based market simulation models [LeBaron, 2006] to modern LLM-powered collaborative frameworks. The key insight driving multi-agent approaches is that financial decision-making in practice involves multiple specialized roles—analysts, traders, risk

managers—whose interactions produce better outcomes than any single agent [Xiao et al., 2024]. Recent multi-agent financial systems draw inspiration from the generative agents framework [Park et al., 2023] and software development multi-agent systems [Qian et al., 2024], adapting these architectures to the financial domain with specialized roles and domain-specific tools.

2.2 Data Sources and Modalities

Financial AI agents operate over a diverse landscape of data sources and modalities. The richness and variety of the data an agent can access fundamentally shapes its perception of the market and the quality of its decisions. This chapter surveys the major data categories used by existing financial agents.

2.2.1 Numerical Market Data

The most fundamental data source for financial agents is numerical market data, including asset prices (open, high, low, close, adjusted close), trading volumes, and derived technical indicators such as moving averages, RSI, MACD, and Bollinger Bands. Nearly all trading agents ingest this data as their primary input. In DRL-based agents such as those built on FinRL [Liu et al., 2022b], the state representation typically consists of price features and technical indicators organized as a vector or matrix over a lookback window. LLM-based agents such as FinMem [Yu et al., 2023] and FinAgent [Zhang et al., 2024] additionally convert numerical data into textual summaries or structured prompts that can be processed by language models.

2.2.2 Textual Financial Data

Textual data constitutes a major advantage of LLM-based agents over traditional quantitative systems. Key textual sources include:

News Articles and Headlines. Financial news is widely used for sentiment analysis and event detection. Agents such as TradingAgents [Xiao et al., 2024] employ dedicated news analyst agents that process and summarize relevant articles. FinGPT [Yang et al., 2023] fine-tunes LLMs on financial news for sentiment-enhanced stock movement prediction.

Financial Reports. Regulatory filings such as SEC Form 10-K and 10-Q provide detailed company financials, management discussions, and risk disclosures. Kim et al. [2024] demonstrated that LLMs can analyze financial statements with accuracy rivaling professional analysts. Trading agents like FinMem [Yu et al., 2023] and TradingGPT [Li et al., 2023] incorporate financial reports to enrich their memory and reasoning.

Social Media and Forums. Platforms such as Twitter/X, Reddit (e.g., r/wallstreetbets), and StockTwits provide real-time retail investor sentiment. Multi-agent systems often include dedicated sentiment analyst agents that process social media feeds [Xiao et al., 2024].

Analyst Reports. Professional analyst reports and investment research provide high-quality expert opinions. FinAgent [Zhang et al., 2024] incorporates expert guidance from platforms such as SeekingAlpha as a key input to its decision-making module.

2.2.3 Visual and Chart Data

Financial charts encode important pattern information that is not easily captured by numerical features alone. FinAgent [Zhang et al., 2024] pioneered the integration of visual data in LLM-based trading agents, incorporating Kline charts and trading charts alongside numerical and textual data. Using multimodal LLMs such as GPT-4V, FinAgent demonstrated that visual inputs can significantly improve trading performance compared to text-and-number-only baselines. Despite this promise, visual financial data remains underexplored relative to text and numerical modali-

ties, partly because most LLMs have not been specifically trained on financial chart interpretation.

2.2.4 Alternative Data

An emerging trend is the incorporation of alternative data sources that go beyond traditional financial data. These include insider trading records and institutional holdings, macroeconomic indicators and central bank communications, supply chain data and satellite imagery, and earnings call audio transcripts. TradingAgents [Xiao et al., 2024] uses insider transaction data and institutional holdings alongside traditional price and news data. The integration of such heterogeneous data sources is an area of active development, particularly as multi-modal foundation models become more capable.

2.2.5 Data Integration Challenges

Financial agents face several data integration challenges. First, different data sources arrive at different frequencies—prices at millisecond to daily intervals, news irregularly, and financial reports quarterly. Agents must handle this temporal misalignment. Second, data quality varies significantly across sources, with social media data being particularly noisy. Third, the volume of available data often exceeds the context window of LLMs, requiring agents to implement summarization, filtering, or retrieval-augmented generation strategies to select the most relevant information.

2.3 Agent Design

This chapter presents a taxonomy of agent design patterns observed in the financial AI literature. We categorize existing work by architectural paradigm rather than by financial application, focusing on the structural choices that define how agents perceive, reason, remember, and act.

2.3.1 Single-Agent Architectures

DRL-Based Agents The simplest agent design is a single DRL agent that directly maps market observations to trading actions. These agents formulate trading as a Markov Decision Process (MDP) where the state includes market features, the action space encompasses buy/hold/sell decisions with optional position sizing, and the reward reflects trading performance. The FinRL library [Liu et al., 2022b,a] standardized this paradigm with a three-layer architecture separating market environment, agent training, and application layers, supporting algorithms including PPO, A2C, DDPG, SAC, and TD3. Zhang et al. [2020] applied deep reinforcement learning with attention mechanisms, and Sun et al. [2021] provided a comprehensive survey of RL for quantitative trading. The key limitation of DRL-based agents is their reliance on numerical state representations, which makes it difficult to incorporate unstructured information such as news or financial reports without extensive feature engineering.

LLM-Based Single Agents LLM-based single agents address the limitation above by using language models as the core reasoning engine. In this design, all financial data—numerical, textual, and sometimes visual—is converted into prompts that the LLM processes to generate trading decisions. The agent typically follows a perceive-reason-act loop: it ingests current market information, reasons about it (potentially using chain-of-thought), and outputs a trading action or signal. FinGPT [Yang et al., 2023] represents a foundational approach, providing open-source financial LLMs fine-tuned for sentiment analysis and trading signal generation. Rather than building a full agent loop, FinGPT focuses on the LLM backbone, enabling downstream agent applications. Lopez-Lira and Tang [2023] demonstrated that even zero-shot LLM predictions based on news headlines can generate profitable trading signals, validating the potential of language models as financial reasoning engines.

2.3.2 Memory-Augmented Agents

A critical challenge for financial agents is maintaining relevant context over time. Memory-augmented agent designs address this by incorporating explicit memory structures that store, organize, and retrieve past observations and reflections.

FinMem [Yu et al., 2023] introduced a layered memory architecture inspired by human cognitive structures. The agent maintains three memory tiers: working memory for immediate context, episodic memory for recent events, and semantic memory for long-term knowledge. Raw inputs such as daily news and financial reports are summarized into memories, and upon new observations, relevant memories are retrieved based on recency, relevancy, and importance scores. This design allows the agent to evolve its knowledge over time and react to new market signals with historical context.

FinAgent [Zhang et al., 2024] extended the memory-augmented paradigm with a dual-level mechanism. It stores not only raw observations but also learned lessons from past trading experiences, creating a self-improving feedback loop. The diversified retrieval operation assigns specific retrieval types to historical information, enabling the agent to access past market intelligence from multiple perspectives and senses.

The memory-augmented design pattern is particularly important in finance because market regimes change over time, and an agent's ability to recall and reason about analogous historical situations can significantly improve decision quality.

2.3.3 Tool-Augmented Agents

Tool-augmented agents extend the base LLM with access to external tools and APIs, enabling capabilities that pure language generation cannot provide. In the financial context, tools may include real-time market data APIs, technical indicator calculators, backtesting engines, web search for current news, code execution environments for quantitative analysis, and order execution interfaces.

FinAgent [Zhang et al., 2024] exemplifies this pattern by integrating established

trading strategies and expert financial tools with the LLM's reasoning. The agent can invoke technical analysis tools, query market data APIs, and execute code to process numerical data that would be unreliable if handled through pure language generation.

QuantAgent [Wang et al., 2024b] takes tool augmentation further by using LLMs not for direct trading but for generating alpha factors—quantitative signals that predict asset returns. The system operates through an inner-loop and outer-loop architecture: the inner loop uses the LLM to generate candidate alpha factor expressions, while the outer loop validates them against historical data using backtesting tools. This design leverages the LLM's creative reasoning for hypothesis generation while relying on quantitative tools for rigorous validation.

Tool augmentation is particularly valuable in finance because financial reasoning often requires precise numerical computation that LLMs handle poorly, and because real-time market data access is essential for actionable decisions.

2.3.4 Reasoning-and-Reflection Agents

Reasoning-and-reflection agents incorporate explicit mechanisms for the agent to evaluate and improve its own reasoning process. This design pattern goes beyond simple chain-of-thought prompting by adding iterative self-critique, reflection on past decisions, and adaptive strategy refinement.

FinAgent [Zhang et al., 2024] implements a dual-level reflection mechanism. At the first level, the agent reflects on individual trading decisions by analyzing whether the outcome matched its reasoning. At the second level, it performs meta-reflection on its overall strategy, identifying systematic biases or recurring errors. These reflections are stored in memory and influence future decisions.

FinMem [Yu et al., 2023] incorporates reflection through its memory architecture: when new observations arrive, the agent generates reflections that synthesize recent memories with new information. Both memories and reflections are stored and retrieved during subsequent decision-making, creating a cycle of experience-driven

improvement.

The reasoning-and-reflection pattern is critical in finance because markets are non-stationary, and strategies that worked in one regime may fail in another. An agent's ability to recognize when its assumptions are no longer valid and adapt accordingly is essential for sustained performance.

2.3.5 Multi-Agent Collaborative Frameworks

Multi-agent designs assign different specialized roles to separate LLM agents that collaborate to produce decisions. This pattern mirrors the organizational structure of real-world trading firms and investment institutions, where analysts, researchers, traders, and risk managers each contribute specialized expertise.

Role-Specialized Agent Teams TradingAgents [Xiao et al., 2024] proposed a framework directly inspired by professional trading firms, featuring agents in specialized roles: fundamental analysts who assess company financials, sentiment analysts who gauge market mood from social media, technical analysts who identify price patterns, news analysts who evaluate macroeconomic developments, and traders who synthesize all inputs to make final decisions. The framework also includes a risk management team that monitors portfolio exposure.

HedgeAgents [Li et al., 2025] introduced a coordinated multi-agent system with a central fund manager agent orchestrating multiple specialized hedging expert agents, each focusing on different asset classes. This hierarchical design enables efficient division of labor while maintaining overall portfolio coherence.

Debate-Driven Agent Architectures A distinctive variant of multi-agent design uses structured debate between agents to improve reasoning quality. In TradingAgents [Xiao et al., 2024], Bull and Bear researcher agents independently assess market conditions from optimistic and pessimistic perspectives, then engage in a structured

debate. The trader agent observes this debate and synthesizes the opposing arguments to make a more balanced decision. Xing [2025] applied heterogeneous LLM agent debate to financial sentiment analysis, where agents with different biases or perspectives analyze the same news and reach more robust classifications through argumentation. This approach helps mitigate the tendency of individual LLMs to exhibit overconfident or biased sentiment predictions.

Information-Distillation Agent Networks FINCON [Yu et al., 2024] proposed a multi-agent system with conceptual verbal reinforcement, addressing the challenge of information overload in financial decision-making. Rather than passing all raw data between agents, FINCON implements an information distillation process where each agent compresses and refines information before passing it to downstream agents. This addresses a key weakness of naive multi-agent systems—the "telephone effect" where details are lost and states become corrupted as information passes through multiple agents.

2.3.6 Hybrid Agent Designs

In practice, many state-of-the-art financial agents combine multiple design patterns. FinAgent [Zhang et al., 2024] is simultaneously memory-augmented, tool-augmented, reasoning-and-reflection-enabled, and multimodal. TradingAgents [Xiao et al., 2024] combines multi-agent collaboration with role specialization and debate mechanisms. This trend toward hybrid designs reflects the inherent complexity of financial decision-making, which demands diverse capabilities that no single design pattern can fully address.

An important design consideration in hybrid systems is the coordination overhead. As agents become more complex, the computational cost of inference increases, the number of potential failure modes grows, and the system becomes harder to debug and interpret. Balancing architectural sophistication with practical deployability remains

an open challenge.

2.4 Applications

This chapter organizes the literature by financial application domain, examining how the agent design patterns described in Section 2.3 are deployed to address specific financial tasks.

2.4.1 Trading

Trading is the most extensively studied application for financial AI agents. The goal is to generate buy/sell/hold signals or execute trades that maximize risk-adjusted returns.

Stock Trading Stock trading has served as the primary testbed for both DRL-based and LLM-based agents. FinRL [Liu et al., 2022b,a] demonstrated that DRL agents (PPO, DDPG, A2C) can outperform traditional strategies on Dow Jones 30 constituents and S&P 500 stocks. LLM-based agents have extended this by incorporating news and sentiment: FinMem [Yu et al., 2023] showed that layered memory with character design improves trading decisions on individual stocks, while FinAgent [Zhang et al., 2024] achieved an average 36% improvement in profitability across six datasets by integrating multimodal data. Multi-agent systems have pushed performance further. TradingAgents [Xiao et al., 2024] demonstrated significant improvements in cumulative returns, Sharpe ratio, and maximum drawdown by simulating the collaborative dynamics of a professional trading firm. FINCON [Yu et al., 2024] showed superior performance over both DRL agents and single LLM agents through its information distillation approach.

Cryptocurrency Trading Cryptocurrency markets, characterized by high volatility, 24/7 operation, and strong social media influence, represent a distinct challenge

for trading agents. FinAgent [Zhang et al., 2024] and FinRL [Liu et al., 2022b] both include cryptocurrency datasets in their evaluations. The unique dynamics of crypto markets—including the importance of on-chain data, social media hype cycles, and regulatory uncertainty—make them a particularly interesting testbed for agent adaptability.

Algorithmic and High-Frequency Trading While most LLM-based agents operate at daily or intraday frequencies, algorithmic and high-frequency trading (HFT) remain primarily the domain of DRL agents and traditional quantitative systems due to the latency constraints of LLM inference. However, LLM agents are increasingly used for longer-horizon components of algorithmic strategies, such as alpha factor generation [Wang et al., 2024b] and regime detection, while faster numerical models handle execution.

2.4.2 Portfolio Management

Portfolio management extends beyond individual asset trading to the construction and rebalancing of multi-asset portfolios.

DRL-based portfolio management agents optimize asset allocation by treating the portfolio weight vector as the action space. FinRL [Liu et al., 2022b] supports portfolio allocation across multiple stocks with algorithms that learn to balance return and risk. LLM-based approaches to portfolio management are emerging: FINCON [Yu et al., 2024] evaluated its multi-agent system on portfolio management tasks comparing against Markowitz mean-variance optimization and equal-weighted strategies. HedgeAgents [Li et al., 2025] addressed multi-asset hedging through a hierarchical multi-agent design where specialized agents manage different asset classes while a central manager ensures overall portfolio coherence. This mirrors real-world asset management organizations where sector specialists report to a chief investment officer.

2.4.3 Risk Management and Fraud Detection

Fraud Detection Traditional fraud detection relies on rule-based filters and supervised classifiers. Agent-based approaches enhance these with autonomous monitoring, adaptive learning, and multi-step reasoning [Hilal et al., 2022]. LLM-powered fraud detection agents can process unstructured data such as customer communications and news alongside structured transaction data, providing a more holistic view of potential fraud patterns [Dong et al., 2025]. The agentic approach enables systems that dynamically adjust detection thresholds, correlate suspicious activities across accounts, and generate human-readable explanations for flagged transactions.

Credit Risk Assessment AI agents for credit risk go beyond traditional scoring by incorporating dynamic information gathering and multi-step reasoning. LLM-based agents can analyze financial statements, assess qualitative risk factors from management discussions, and integrate macroeconomic indicators [Kim et al., 2024]. The agentic design enables systems that actively seek additional information when initial assessments are uncertain and provide transparent reasoning chains for regulatory compliance.

Market Risk Monitoring Multi-agent systems have been applied to market risk monitoring, where different agents track different risk dimensions—market risk, credit risk, liquidity risk, and operational risk—and coordinate their assessments. The risk management team in TradingAgents [Xiao et al., 2024] exemplifies this approach, monitoring portfolio exposure and intervening when risk thresholds are breached.

2.4.4 Financial Analysis and Advisory

Financial Report Analysis LLM agents have demonstrated strong capabilities in analyzing financial reports. Kim et al. [2024] showed that GPT-4 can generate earnings predictions rivaling professional analysts. The agent-based extension in-

volves systems that autonomously collect filings, extract key metrics, compare against consensus expectations, and generate investment recommendations with supporting evidence.

Sentiment-Driven Analysis Sentiment analysis agents process news, social media, and analyst opinions to gauge market sentiment. FinGPT [Yang et al., 2023] provides the LLM backbone for sentiment-enhanced analysis. MarketSenseAI [Fatouros et al., 2025] integrated multiple information sources for stock analysis through LLM agents. Multi-agent debate has been applied to sentiment analysis, where agents with different perspectives reach more robust conclusions through structured argumentation [Xing, 2025].

Robo-Advisory AI agents for financial advisory combine financial knowledge with personalized client profiling. These systems assess risk tolerance, financial goals, and constraints, then generate and monitor investment plans. The challenge lies in balancing automation with regulatory requirements, which in many jurisdictions mandate human oversight for personalized advice. Conversational AI agents in banking handle routine inquiries, explain portfolio performance, and escalate complex situations to human advisors.

2.4.5 Regulatory Compliance and Reporting

Compliance Monitoring Financial institutions must comply with extensive regulatory frameworks including AML rules, data privacy standards, and reporting obligations [Arner et al., 2017]. LLM-based compliance agents can interpret regulatory text, map requirements to internal controls, and flag gaps. Their advantage over rule-based systems lies in handling ambiguous situations, adapting to regulatory changes, and providing natural language explanations.

Automated Financial Reporting Agentic AI systems automate financial reporting workflows including data collection, reconciliation, analysis, and report generation [Nie et al., 2024]. Multi-agent architectures assign specialized roles—data gathering, validation, analysis, and drafting agents—that collaborate to produce comprehensive reports.

Anti-Money Laundering AML is a high-value application for AI agents due to the volume of transactions requiring monitoring and the complexity of laundering schemes. Agent-based AML systems trace funds across accounts and institutions, identify layering patterns, and generate suspicious activity reports with evidence chains.

2.5 Benchmarks and Evaluation

2.5.1 Trading Performance Metrics

Financial AI agents are typically evaluated using standard quantitative finance metrics. Cumulative Return (CR) measures total return over the evaluation period. Sharpe Ratio (SR) captures risk-adjusted return as excess return per unit of volatility. Maximum Drawdown (MDD) measures the largest peak-to-trough decline, quantifying downside risk. Additional metrics include the Sortino Ratio (considering only downside deviation) and the Calmar Ratio (annualized return divided by maximum drawdown).

2.5.2 Financial NLP Benchmarks

For agents that process textual data, several benchmarks evaluate component capabilities. FinBen [Xie et al., 2024] provides a holistic benchmark covering sentiment analysis, named entity recognition, question answering, and numerical reasoning over

financial data. FinanceBench [Islam et al., 2023] focuses on question answering over real financial documents including SEC filings.

2.5.3 Agent-Level Benchmarks

A critical gap in the current literature is the lack of comprehensive agent-level benchmarks that evaluate end-to-end task completion in realistic financial scenarios. While individual systems are evaluated on their specific tasks, standardized benchmarks that enable fair comparison across different agent architectures remain scarce. Emerging efforts such as FinAgentBench [Choi et al., 2025] aim to address this by evaluating agents on multi-step financial tasks that require information gathering, reasoning, and action.

2.5.4 Evaluation Challenges

Several challenges complicate the evaluation of financial AI agents. First, backtesting on historical data may overestimate performance due to look-ahead bias, survivorship bias, and the inability to account for market impact. Second, live trading evaluation is expensive and risky. Third, comparing agents across different papers is difficult due to variations in datasets, time periods, transaction cost assumptions, and baseline configurations [Liu et al., 2022a]. Fourth, many LLM-based agents use proprietary models (e.g., GPT-4), making exact reproduction challenging. Fifth, financial markets are non-stationary, so strong historical performance does not guarantee future results.

Chapter 3

Research Methodology

3.1 Gap Analysis

Existing work on LLM-based financial reasoning falls into two camps that address different parts of the problem. The first camp, exemplified by TradingAgents [Xiao et al., 2024], FinCon [Yu et al., 2024], and FinGPT [Yang et al., 2023], builds multi-agent systems that generate equity-level trading signals from news sentiment, technical indicators, and fundamental data. These systems operate primarily at the individual asset level; they do not, for instance, trace how a Federal Reserve rate decision propagates through the credit channel to affect corporate bond spreads or equity risk premia. The second camp, including EconAgent [Li et al., 2024] and Lopez-Lira and Tang [2023], applies LLMs to macroeconomic simulation or forecasting but is not directly connected to calibrated cross-asset allocation signals. MarketSenseAI [Fatouros et al., 2025], while primarily an equity analysis agent for S&P 100 stocks, similarly operates without explicit macro transmission-mechanism reasoning or cross-asset calibration.

Three gaps motivate the design of Macro-MOSAIC. First, we are not aware of a system that generates **explicit, auditable multi-school parallel reasoning chains** where structurally distinct economic frameworks (Keynesian, Austrian, Monetarist) independently analyze the same macro event and produce divergent causal

chains that are then formally aggregated. FOMC In Silico [Kazinnik and Sinclair, 2025] comes closest by assigning dovish/hawkish personas to FOMC member agents, but these represent individual viewpoints on a single policy variable (the federal funds rate) rather than structurally distinct theoretical frameworks applied to general macro events. Second, **historical analogy retrieval at the macroeconomic episode level**—combining semantic similarity over event narratives with structural similarity over macro state vectors—has not, to our knowledge, been addressed in the systems we surveyed. Existing financial RAG systems [Zhang et al., 2024, Lithgow-Serrano et al., 2025] retrieve similar price patterns or sentiment-level few-shot examples, not structured narratives of comparable macro crises with documented transmission mechanisms and multi-asset outcomes. Third, the literature lacks a **verified end-to-end pipeline** connecting qualitative macro event interpretation through calibrated probability distributions to cross-asset allocation signals, evaluated against both reasoning benchmarks and realized financial outcomes.

3.2 Problem Formulation

Given a macro event e (e.g., a central bank decision, geopolitical shock, or fiscal policy announcement) and the prevailing macro state $\mathbf{s}$ (interest rates, inflation, credit conditions, etc.), the system must produce three outputs:

1. **Reasoning chains** $\{C_k(e, \mathbf{s})\}_{k=1}^{K}$: For each of K economic schools, a structured causal chain tracing the event through specific transmission mechanisms to its expected impact on asset classes.

2. **Calibrated scenario probabilities** $\{(S_j, p_j)\}_{j=1}^{J}$: A discrete set of mutually exclusive scenarios with probabilities satisfying $\sum_j p_j = 1$, where the predicted probability distribution is well-calibrated (i.e., events predicted with probability p occur with empirical frequency p).

3. **Asset allocation signals $\mathbf{w} \in \mathbb{R}^N$**: A vector of signed position weights across N asset classes, derived from the probability-weighted conditional asset return expectations.

The system is evaluated on: (a) reasoning quality, measured against causal economic benchmarks (EconLogicQA [Quan and Liu, 2024], EconCausal [Lee et al., 2025]); (b) calibration quality, measured by Brier Score and Expected Calibration Error against event resolutions; and (c) signal quality, measured by Sharpe ratio, maximum drawdown, and directional accuracy on out-of-sample walk-forward backtests.

3.3 Macro-MOSAIC: System Architecture

We propose **Macro-MOSAIC** (Mixture-Of-Schools Analogy-Informed Chains), a four-stage pipeline that bridges qualitative macroeconomic reasoning and quantitative asset allocation through explicit causal chain construction.

3.3.1 Stage 1: Historical Analogy Retrieval

When a new macro event e arrives, we retrieve the m most relevant historical episodes from a curated database of 141 macro episodes spanning 1971–2025 (see Section 4.1). Each episode is a structured record containing the triggering event, prevailing macro context (35 FRED indicators), tagged transmission channels, multi-horizon asset reactions across 22 instruments, and school-specific interpretive narratives.

Retrieval operates via a dual-channel scoring mechanism. The *semantic channel* computes cosine similarity between the embedding of the incoming event description and the pre-computed embeddings of each episode's narrative text. The *structural channel* computes Mahalanobis distance between the current macro state vector $\mathbf{s}$ and each episode's recorded macro context $\mathbf{s}_i$, capturing similarity in the economic regime (rate environment, inflation level, credit conditions) rather than surface-level

narrative similarity. The final retrieval score is a weighted combination:

$$\text{score}(e, e_i) = \alpha \cdot \cos_\text{sim}(\mathbf{v}_e, \mathbf{v}_{e_i}) + (1 - \alpha) \cdot \exp\big(-d_M(\mathbf{s}, \mathbf{s}_i)\big) \tag{3.1}$$

where α is a tunable parameter and d_M denotes the Mahalanobis distance. The top-m episodes, along with their historical asset reactions, are injected into the reasoning context for Stage 2.

3.3.2 Stage 2: Multi-School Causal Chain Generation

Each retrieved analogy is interpreted through $K = 3$ distinct economic school perspectives (Keynesian, Austrian, and Monetarist), implemented as separate LoRA adapters [Hu et al., 2022] fine-tuned on school-specific corpora. These three schools were selected for three reasons. First, they represent structurally distinct theoretical frameworks with genuinely different transmission mechanism logic: Keynesian analysis centers on aggregate demand and fiscal multipliers, Austrian analysis on capital structure distortions and malinvestment, and Monetarist analysis on money supply dynamics and expectations—producing divergent causal chains for the same macro event. Second, these three schools are prominently represented in the academic and policy discourse relevant to macro-financial analysis: Keynesian frameworks appear in central bank communications and fiscal policy debates, Austrian frameworks provide a contrasting supply-side and capital-structure lens, and Monetarist frameworks anchor inflation and monetary transmission analysis. Their perspectives are therefore directly relevant to cross-asset signal generation even if none can be characterised as universally dominant in institutional practice. Third, they have sufficient published literature (NBER working papers filterable by JEL codes E12, B53, E41/E51) to construct school-specific fine-tuning corpora of adequate size and quality. Post-Keynesian and Institutionalist perspectives, while intellectually valuable, were excluded due to smaller and less standardized corpora, which we leave as a direction for future exten-

sion.

The expected benefit of aggregating diverse school-specific forecasts rests on a standard ensemble learning argument: if individual school models make *different* systematic errors—for instance, Keynesian frameworks tend to underweight supply-side transmission channels while Austrian frameworks tend to underweight demand-multiplier effects—their errors will partially offset under aggregation, reducing the ensemble's overall systematic bias. This argument requires two empirical conditions to hold. First, each school must achieve above-chance directional accuracy on at least a subset of event categories, so that its contribution to the ensemble is informative rather than noise. Second, the school-specific outputs must be genuinely distinct rather than stylistic variations of a single mainstream analysis. Our preliminary results in Section 4.3 provide partial evidence on both conditions: Table 4.7 (Section 4.3.5) shows that the Monetarist adapter produces the largest consistency gains on inflation and employment events while gains are smallest for geopolitical shocks, suggesting category-level complementarity across schools; Table 4.4 shows that fine-tuned adapters achieve a mean inter-school Jaccard distance of 0.853 versus 0.778 for the base model, confirming signal-level diversity. Whether these Stage 2 properties translate into improved calibration scores and risk-adjusted returns in the end-to-end pipeline remains an open empirical question, which the ablation experiments in Section 5.4 are specifically designed to isolate and test.

- **Keynesian adapter**: Fine-tuned on New Keynesian macro literature (IS-LM/AD-AS framing, multiplier analysis, output gap reasoning, liquidity trap dynamics).

- **Austrian adapter**: Fine-tuned on Austrian school texts (ABCT malinvestment logic, Cantillon effects, natural rate deviation analysis, liquidationist reasoning).

- **Monetarist adapter**: Fine-tuned on monetarist literature (quantity theory

transmission, velocity analysis, expectations-augmented Phillips Curve, rules-vs-discretion framing).

Each adapter takes as input the event description, current macro state, and retrieved historical analogies, and produces a structured causal chain in a fixed schema:

$$C_k(e, \mathbf{s}) = \Big[\text{Event} \xrightarrow{\text{mechanism}_k} \text{Intermediate Effects} \xrightarrow{\text{channel}_k} \text{Asset Impacts}_k,\ \hat{p}_k\Big] \tag{3.2}$$

where $\hat{p}_k$ is the adapter's raw confidence estimate for its predicted scenario. Each chain explicitly names the transmission mechanism (credit channel, exchange rate channel, expectations channel, etc.) and maps to directional asset class impacts, ensuring interpretability and auditability.

The LoRA adapters are trained on school-specific corpora curated from NBER Working Papers filtered by JEL classification codes (E12 for Keynesian, B53 for Austrian, E41/E51 for Monetarist), supplemented with canonical textbook passages and FOMC communication data annotated for hawkish/dovish stance [Shah et al., 2023].

3.3.3 Stage 3: Bayesian Aggregation and Calibration

The K school-specific outputs are aggregated into a unified probability distribution via Bayesian model averaging. Rather than fixed equal weighting, each school receives a dynamic weight w_k based on its historical calibration performance within the detected macro regime:

$$w_k^{(t)} \propto \exp\Big(-\lambda \cdot \text{BS}_k^{(\text{regime}(t))}\Big) \tag{3.3}$$

where $\text{BS}_k^{(\text{regime})}$ is school k's rolling Brier Score on resolved predictions within the current macro regime (e.g., tightening cycle, crisis, expansion), and λ is a temperature parameter controlling the sharpness of weighting. Regimes are detected via a Hidden

Markov Model fitted on returns and volatility.

The weighted aggregate probability $\hat{p}_{\text{agg}} = \sum_k w_k \hat{p}_k$ is then passed through a calibration layer trained on historical prediction–resolution pairs sourced from resolved Polymarket and Metaculus questions. We adopt Platt scaling [Platt, 1999] as the default calibration method, to be compared against isotonic regression and temperature scaling in ablation experiments (Section 5.2):

$$p_{\text{cal}} = \sigma\big(a \cdot \text{logit}(\hat{p}_{\text{agg}}) + b\big) \tag{3.4}$$

where σ is the sigmoid function and (a, b) are learned parameters. This two-stage process—regime-conditional school weighting followed by Platt scaling calibration—addresses both the model selection problem (which school is most accurate in this regime?) and the overconfidence problem (LLMs systematically produce poorly calibrated probabilities, as documented by Nel [2025]).

Calibration transfer assumption. We acknowledge that prediction-market questions (Polymarket, Metaculus) differ from macro-to-asset-return forecasts in resolution horizon, phrasing, and outcome granularity. The calibration layer is therefore best interpreted as correcting systematic overconfidence in the model's probability outputs rather than as a guarantee of calibration on financial return targets. Whether the learned (a, b) parameters generalise to asset-return scenarios will be tested empirically in Section 5.2 via Brier Score comparisons between calibrated and uncalibrated signals on held-out macro events with known market outcomes.

3.3.4 Stage 4: Asset Allocation Signal Generation

For each scenario S_j with calibrated probability p_j, the system produces conditional asset return forecasts $\hat{\mathbf{r}}_j \in \mathbb{R}^N$ informed by the causal chains and historical analogy

outcomes. The unconditional expected return vector is:

$$\mathbb{E}[\mathbf{r}] = \sum_j p_j \cdot \hat{\mathbf{r}}_j \tag{3.5}$$

The allocation weight for each asset is computed via a fractional Kelly criterion:

$$w_n = f \cdot \frac{\mathbb{E}[r_n]}{\hat{\sigma}_n^2} \tag{3.6}$$

where $\hat{\sigma}_n^2 = \sum_j p_j \hat{r}_{j,n}^2 - \big(\mathbb{E}[r_n]\big)^2$ is the scenario-implied variance, and $f \in (0, 1]$ is a fractional Kelly parameter (typically $f = 0.25$) that controls aggressiveness. Positions are subject to maximum absolute weight constraints ($|w_n| \leq w_{\max}$) and net exposure limits.

The final output for each event is a structured record containing: the retrieved historical analogies, three school-specific causal chains, the calibrated scenario probability distribution, and the resulting cross-asset allocation signal with per-asset conviction scores.

Chapter 4

Preliminary Results

4.1 Episode Database Construction

We construct a structured historical macro episode database covering 54 years of major macroeconomic events, designed to serve as the retrieval corpus for Stage 1 of the Macro-MOSAIC pipeline. To our knowledge, no existing publicly available dataset simultaneously links macro event narratives, school-specific economic interpretations, quantitative macro state snapshots, and multi-asset market reactions within a unified schema; the closest prior work addresses only subsets of these components in isolation. Table 4.1 summarizes the database.

Coverage. The 141 episodes are organized into eight categories: central bank policy decisions (25), banking and credit crises (20), geopolitical shocks (17), commodity shocks (17), currency and exchange rate crises (16), inflation and stagflation events (16), fiscal policy events (16), and pandemic or black swan events (14). The temporal distribution reflects both the increasing frequency of macro-financial events in recent decades and the greater availability of high-frequency data: 4 episodes from the 1970s, 8 from the 1980s, 15 from the 1990s, 18 from the 2000s, 44 from the 2010s, and 52 from the 2020s.

Table 4.1: Episode database summary statistics.

Attribute	**Value**
Total episodes	141
Temporal coverage	1971–2025
Event categories	8
Unique tags	453
Transmission channels	19
Quantitative layers	
FRED macro indicators per episode	35
Financial instruments tracked	22
Return horizons per instrument	5
Total asset reaction data points	13,571
Qualitative layers	
Episodes with narrative summaries	117
Episodes with 3 school interpretations	117
Mean narrative length (chars)	3,036
Mean school interpretation length (chars)	935

Quantitative layers. Each episode is anchored to a specific trigger date and contains three quantitative layers. The *macro context snapshot* records 35 economic indicators from the Federal Reserve Economic Data (FRED) system at the trigger date, covering interest rates (federal funds rate, 2Y/5Y/10Y/30Y Treasury yields, yield curve spreads), inflation (CPI, core CPI, core PCE, breakeven inflation, Michigan expectations), employment (unemployment rate, nonfarm payrolls, initial claims), money and credit (M2, bank credit, senior loan officer survey), financial conditions (high-yield OAS, TED spread, St. Louis Financial Stress Index), fiscal position (federal debt, surplus/deficit), and housing (Case-Shiller index, housing starts). The *trigger-date price snapshot* records closing levels for 22 financial instruments across equities (S&P 500, DJIA, NASDAQ, FTSE 100, Nikkei 225, Shanghai Composite, EEM), fixed income (10Y/5Y/30Y/3M yields, TLT, HYG), currencies (DXY, USD/JPY, EUR/USD, GBP/USD), commodities (gold, silver, WTI, Brent), and volatility (VIX). The *asset reaction panel* records realized returns (or level changes for yields and VIX) at five horizons post-trigger: 1 week, 1 month, 3 months, 6 months,

and 12 months, yielding 13,571 individual data points across all episodes. This count is derived as $141 \times 22 \times 5 = 15{,}510$ theoretical cells minus 1,939 missing observations arising from two sources: (i) instruments not yet in existence at the time of early episodes (e.g., EEM was launched in 2003, HYG in 2007, and breakeven inflation series begin in the mid-1990s), for which no value is imputed; and (ii) incomplete return windows for episodes occurring close to the data extraction date, where some forward horizons have not yet resolved. Each return is measured from the market close on the trigger date to the market close at the corresponding horizon, using adjusted closing prices from Bloomberg and Yahoo Finance cross-validated against FRED where applicable. For yield instruments and VIX, we record level changes in basis points and index points respectively rather than log returns. Event windows are non-overlapping at the 1-week horizon by construction (no two trigger dates fall within the same calendar week); at longer horizons, windows may overlap and returns are treated as independent observations for analysis purposes.

Qualitative layers. For 117 of the 141 episodes, we generate two qualitative layers using an LLM (Claude Sonnet). The *narrative summary* (mean length: 3,036 characters) provides a factual account of the event including key dates, market moves, policy responses, and aftermath. The *school-specific interpretations* provide three parallel analyses of the same episode—Keynesian (mean: 935 characters), Austrian (mean: 968 characters), and Monetarist (mean: 905 characters)—each framed exclusively within that school's axiomatic framework and terminology. These interpretations serve two purposes: as *few-shot exemplars* injected into the reasoning context during Stage 2 inference, and as *seed content* whose vocabulary and causal structure informed the construction of the school-specific synthetic trajectory templates used for LoRA adapter training (Section 4.2). Critically, the adapters are not trained directly on these 117 episode interpretations; instead, 500 synthetic trajectories per school are generated from 107 abstract event templates instantiated with randomised macro

Table 4.2: Asset reaction validation against known benchmarks. S&P 500 returns and VIX changes are measured at the 1-month horizon from the trigger date.

Episode	**S&P 500 (1M)**	**ΔVIX (1M)**	**Δ10Y (bp)**
Lehman (Sep 2008)	−16.3%	+23.4	+54.0
COVID crash (Feb 2020)	−24.1%	+36.6	−56.1
Trump tariffs (Apr 2025)	+0.3%	+1.2	+12.6

state snapshots, as described in Section 4.2.1.

Structural metadata. Each episode is annotated with categorical tags (453 unique tags across 583 assignments) and transmission channel labels drawn from a taxonomy of 19 channels. The most frequently tagged channels are confidence (75 episodes), expectations (64), credit (43), liquidity (33), interest rate (30), exchange rate (28), and commodity (28). This channel taxonomy directly maps to the causal chain schema used in Stage 2, enabling structured retrieval: when the system identifies an incoming event as primarily operating through the credit channel, it can preferentially retrieve episodes tagged with the same transmission mechanism.

Validation. We verify the asset reaction data against known historical benchmarks. Table 4.2 reports three representative episodes. The recorded values are consistent with independently reported market data, confirming the integrity of the automated collection pipeline.

4.2 School-Specific Agent Evaluation

We evaluate whether fine-tuning separate LoRA adapters on school-specific agent trajectories produces agents that (1) consistently reason within their designated economic framework, (2) adopt the structural behavior pattern of tool-augmented agents, and (3) generate meaningfully divergent analyses across schools. We compare three fine-tuned school agents—Keynesian, Austrian, and Monetarist—against a shared

base model prompted with equivalent school-specific system instructions.

4.2.1 Data

Training data. For each of the three schools, we generate 500 synthetic agent trajectories using a template-based pipeline. Each trajectory consists of a structured multi-step reasoning chain: `[Think]` → `[ToolCall: query_episode_db]` → `[Observe]` → `[Think]` → `[ToolCall: get_macro_data]` → `[Observe]` → `[Think]` → `[Output]`. The output includes a causal reasoning chain grounded in the school's theoretical concepts, a scenario probability distribution, and cross-asset directional signals with conviction scores.

The 500 trajectories per school are generated from 107 unique macroeconomic event templates spanning 14 categories (monetary policy, fiscal policy, inflation data, employment data, geopolitical shocks, trade and tariffs, commodity shocks, market events, economic data surprises, China-specific, emerging market crises, housing, banking stress, and black swan events). This finer-grained categorization extends the eight episode database categories by splitting broad classes (e.g., separating monetary policy into distinct central bank decisions, inflation data, and employment data events) to improve coverage of training scenarios. Each event template is instantiated with varying macroeconomic context snapshots (randomized Fed funds rate, CPI, unemployment, VIX, DXY, etc.) to ensure the model learns the reasoning *pattern* rather than memorizing specific event-response pairs. Because training trajectories are generated from abstract event templates with randomized macro state snapshots rather than from specific historical episode records, the fine-tuning corpora contain no instance-level information about the 2023–2025 events used in out-of-sample evaluation, precluding data leakage from the episode database into the backtesting period. Crucially, each school's trajectories differ in three dimensions: (a) the core theoretical concepts invoked (e.g., *multiplier effect* for Keynesian vs. *malinvestment* for Austrian vs. *M2 money supply* for Monetarist), (b) the transmission mechanisms cited, and (c)

the directional asset signals produced.

Test data. We construct a held-out evaluation set of 96 novel macroeconomic events that are *structurally distinct* from the training templates. These include multi-factor contradictory events (e.g., "The Fed cuts rates by 50bp while simultaneously announcing a $500 billion QE program, but the 10-year yield rises 30bp on the announcement"), unprecedented policy actions (e.g., "BRICS+ launches a gold-backed digital settlement currency"), and complex data surprises (e.g., "US Q3 GDP comes in at +3.8% annualized, but GDI shows −0.5%, the largest GDP–GDI divergence on record"). Events are labeled by difficulty (*medium*: 42, *hard*: 47, *very hard*: 7) and by expected inter-school divergence.

4.2.2 Model and Training

We fine-tune three LoRA adapters on a shared Qwen2.5-7B-Instruct base model [Qwen Team, 2025]. Each adapter is trained independently on its school's 500-trajectory dataset. We use 4-bit NormalFloat quantization [Dettmers et al., 2023] with double quantization to fit the 7B model within 24GB GPU memory. The LoRA configuration uses rank $r = 32$, $\alpha = 64$, and targets all attention and MLP projection matrices (`q_proj`, `k_proj`, `v_proj`, `o_proj`, `gate_proj`, `up_proj`, `down_proj`), yielding approximately 80.7M trainable parameters (1.05% of total). Training runs for 3 epochs with a cosine learning rate schedule (peak 2×10^{-4}), AdamW-8bit optimizer, effective batch size of 16, and gradient checkpointing enabled.

The baseline condition uses the same Qwen2.5-7B-Instruct model without any adapter, prompted with a school-specific system instruction describing the school's core principles and requesting the same output format (causal chain, scenario probabilities, asset signals).

4.2.3 Evaluation Metrics

We evaluate on four dimensions:

School consistency score. We define a keyword-based scoring function for each school. For each school s, we maintain three keyword lists: (a) *must-have* core concepts (6–7 per school, e.g., {*aggregate demand*, *multiplier*, *output gap*, *fiscal*, *effective demand*, *consumption*, *investment*} for Keynesian); (b) *strong-signal* terms that indicate deeper school-specific reasoning (e.g., *liquidity trap*, *IS-LM*, *paradox of thrift*); and (c) *anti-signal* terms from competing schools that indicate theoretical contamination (e.g., the presence of *malinvestment* or *MV=PQ* in a Keynesian response). The consistency score is:

$$C_s = \max\left(0,\ \frac{1}{2} \cdot \frac{|\text{must-have hits}|}{|\text{must-have pool}|} + \frac{1}{2} \cdot \min\left(\frac{|\text{strong hits}|}{5},\ 1\right) - 0.15 \cdot |\text{anti-signal hits}|\right) \tag{4.1}$$

Cross-school contamination. We report the mean number of anti-signal keyword hits per response. Lower values indicate purer school-specific reasoning with less "leakage" from competing frameworks.

Structure completeness. We check for the presence of seven structural elements in each response: `[Think]`, `[ToolCall]`, `[Observe]`, `[Output]`, a causal reasoning chain, scenario probabilities, and asset signals. The completeness score is the fraction of these elements present.

Inter-school diversity. For each test event, we extract the directional asset signals (long/short/neutral) for major asset classes from all school responses and compute the mean pairwise Jaccard distance:

$$D = \frac{1}{\binom{K}{2}} \sum_{i<j} \left(1 - \frac{|S_i \cap S_j|}{|S_i \cup S_j|}\right) \tag{4.2}$$

Table 4.3: School consistency scores for fine-tuned (SFT) vs. base model with prompt-only school instructions, evaluated on 96 held-out events. Win rate denotes the fraction of events on which the SFT adapter outscores the base model.

School	Base	SFT	Δ	Rel. Improv.	SFT Win Rate
Keynesian	0.368 ± 0.163	0.338 ± 0.175	-0.030	-8.1%	39.6%
Austrian	0.146 ± 0.109	0.319 ± 0.187	$+0.173$	$+118.8\%$	75.0%
Monetarist	0.230 ± 0.137	0.528 ± 0.233	$+0.298$	$+129.7\%$	83.3%
Average	0.248	0.395	$+0.147$	$+59.3\%$	65.6%

where S_i is the set of (asset, direction) pairs produced by school i and K is the number of schools. Higher diversity indicates that the schools are generating meaningfully different analytical perspectives, which is a prerequisite for effective ensemble aggregation.

4.3 Results

We report results across all 96 held-out test events. Table 4.3 summarizes the main findings.

4.3.1 School Consistency

Fine-tuning produces substantial gains in school-specific reasoning for the Austrian and Monetarist agents, with more modest results for the Keynesian agent.

The **Monetarist** adapter achieves the largest improvement, with a consistency score of 0.528 compared to 0.230 for the base model (+129.7%). The SFT agent wins on 80 of 96 events (83.3%). Examining the sub-components, we find that the SFT Monetarist agent hits 3.6 out of 6 must-have keywords per response (vs. 2.6 for base) and 2.4 strong-signal keywords (vs. 1.4). The base model rarely produces core Monetarist terms like *MV=PQ*, *quantity theory*, or *long and variable lags* unless explicitly prompted, whereas the fine-tuned agent incorporates these concepts as

natural components of its reasoning chain.

The **Austrian** adapter shows similarly large relative improvement (+118.8%), rising from 0.146 to 0.319, with SFT winning on 75.0% of events. The base model scores particularly low on Austrian must-have terms (0.3 out of 6), indicating that Qwen2.5-7B-Instruct has limited familiarity with heterodox Austrian concepts such as *malinvestment*, *ABCT*, and *capital structure distortion*. Fine-tuning raises must-have hits to 1.9 out of 6, demonstrating successful internalization of these domain-specific concepts.

The **Keynesian** adapter shows a small decrease in the composite consistency score (−8.1%), which warrants careful interpretation. The apparent regression is an artifact of the scoring metric, not a degradation in reasoning quality: the deficit is driven entirely by reduced strong-signal keyword hits, while school purity and output structure both improve substantially. Decomposing the score reveals that the deficit is driven entirely by a reduction in strong-signal hits (2.1 → 0.9), while must-have keyword coverage *improves* (3.0 → 3.4 out of 7), anti-signal contamination drops to zero (0.44 → 0.00), and structured output block presence rises from 59% to 97%. We attribute the strong-signal decline to a fundamental asymmetry between the Keynesian school and the other two: the Qwen2.5-7B-Instruct base model has been pretrained on large volumes of mainstream macroeconomic text in which Keynesian vocabulary is the dominant analytical idiom. The base model therefore produces stylistically Keynesian outputs without fine-tuning, but its fluency is undifferentiated—conflating New Keynesian IS-LM reasoning with Post-Keynesian and Old Keynesian terminology into a single vocabulary pool. Fine-tuning specializes the adapter toward New Keynesian axiomatic structure, causing it to avoid Post-Keynesian strong-signal terms (e.g., *endogenous money*, *Minsky moment*) that fall outside this sub-school's scope—reducing keyword count while improving theoretical coherence. For the downstream pipeline, the properties that matter are output parsability and school purity, not standalone keyword density: the SFT Keynesian adapter's 97% structured output rate (vs. 59%

for base) and perfect anti-signal elimination make it substantially more reliable as an input to the Bayesian aggregation stage than the base model's superficially higher consistency score would suggest.

4.3.2 Cross-School Contamination

Table 4.4 reports the mean number of anti-signal keywords per response—terms from competing schools that appear in a given school's output.

Table 4.4: Cross-school contamination measured by anti-signal keyword hits per response (lower is better). SFT adapters substantially reduce leakage from competing frameworks.

School	**Base**	**SFT**	**Reduction**
Keynesian	0.44	**0.00**	−100%
Austrian	1.21	**0.59**	−51.2%
Monetarist	0.88	**0.02**	−97.7%

The reduction in contamination is striking. The base Keynesian agent occasionally references *Minsky* dynamics or *endogenous money* (Post-Keynesian concepts) and *malinvestment* (Austrian); the SFT agent produces zero anti-signal hits across all 96 events. The base Monetarist agent averages 0.88 anti-signal terms per response—frequently invoking *multiplier effect* (Keynesian) or *Minsky* (Post-Keynesian)—while the SFT agent reduces this to 0.02. The base Austrian agent shows the highest contamination at 1.21 anti-signal terms, often producing mainstream Keynesian language (*output gap*, *aggregate demand*); SFT reduces this by 51.2%.

This finding has direct implications for the downstream Bayesian aggregation layer: if individual school agents produce contaminated outputs that partially overlap with other schools' perspectives, the aggregation step receives redundant rather than complementary signals, reducing the effective diversity of the ensemble.

Table 4.5: Structural completeness scores and sub-component rates (% of responses containing each element).

	Completeness			Output		Scenarios	
School	Base	SFT	Δ	Base	SFT	Base	SFT
Keynesian	0.742	0.693	−0.049	59%	**97%**	93%	91%
Austrian	0.763	0.768	+0.004	81%	**93%**	93%	94%
Monetarist	0.628	**0.717**	+0.089	51%	**99%**	85%	**100%**

4.3.3 Output Structure

The SFT agents show marked improvement in the `[Output]` block adherence, which contains the structured causal chain, scenarios, and signals. The base Monetarist agent produces a recognizable `[Output]` section in only 51% of responses, while the SFT agent does so in 99%. Similar gains appear for Keynesian (59% → 97%) and Austrian (81% → 93%). The Monetarist adapter also achieves 100% scenario probability generation (vs. 85% for base) and 100% signal generation (vs. 91%). These improvements are critical for the end-to-end pipeline, as downstream components (Bayesian aggregation, asset allocation) require parseable structured output.

4.3.4 Inter-School Diversity

The SFT ensemble produces higher inter-school diversity than the base ensemble on 62.5% of test events, with a mean Jaccard distance of 0.853 ± 0.082 compared to 0.778 ± 0.158 for base ($p < 0.01$). The SFT ensemble's lower variance (0.082 vs. 0.158) indicates more *consistently* diverse outputs, whereas the base ensemble occasionally produces near-identical responses across schools.

This result is consistent with a key design prerequisite of Macro-MOSAIC: fine-tuning on school-specific trajectories produces agents that generate meaningfully different analytical perspectives on the same event, rather than minor stylistic variations of a single mainstream analysis. Whether this inter-school diversity translates into improved calibration and financial signal quality remains to be evaluated in Stages 3

and 4.

4.3.5 Analysis by Event Difficulty and Category

Table 4.6: Mean consistency score (three-school average) by event difficulty.

Difficulty	**Base**	**SFT**	Δ
Medium ($n = 42$)	0.241	0.395	+0.154
Hard ($n = 47$)	0.254	0.391	+0.137
Very Hard ($n = 7$)	0.252	0.425	+0.173

Fine-tuning gains are consistent across difficulty levels, and the largest absolute improvement occurs on the most challenging events (+0.173 for *very hard*). These events involve multi-factor contradictions—such as simultaneous deflation and commodity spikes, or failed easing where long-term yields rise despite rate cuts—where school-specific frameworks provide structured interpretive scaffolding that the base model's generic analysis lacks.

Table 4.7 presents results by event category. SFT gains are largest for data-driven events—inflation data (+0.261), economic data (+0.259), and employment data (+0.227)—where the schools' theoretical frameworks offer the most differentiated causal interpretations. For instance, on an unexpected CPI reacceleration, the Monetarist agent attributes the outcome to prior M2 growth with a 12–18 month lag, while the Austrian agent frames it as a consequence of malinvestment-driven supply bottlenecks. Gains are smallest for geopolitical events (−0.002), where all schools tend to converge on similar supply-shock analyses.

4.3.6 Summary of Findings

These preliminary results indicate that LoRA fine-tuning on 500 synthetic agent trajectories per school produces meaningfully differentiated economic reasoning agents at the Stage 2 level. The fine-tuned agents exhibit (1) substantially higher school-

Table 4.7: SFT improvement (Δ in consistency score) by event category, three-school average, sorted by improvement magnitude. Categories with fewer than 5 test events are omitted.

Category	**Δ (SFT − Base)**	**Events**
Inflation data	+0.261	10
Economic data	+0.259	8
Employment data	+0.227	8
Monetary policy	+0.180	15
Market events	+0.170	10
China-specific	+0.168	7
Commodity shocks	+0.117	8
Fiscal policy	+0.084	10
Trade / tariffs	+0.014	7
Geopolitical	−0.002	12

style consistency for non-mainstream frameworks (Austrian +118.8%, Monetarist +129.7%), (2) near-complete elimination of cross-school contamination (Keynesian anti-signal hits reduced from 0.44 to 0.00; Monetarist from 0.88 to 0.02), (3) more reliable structured output generation (Output block presence rising from 51–81% to 93–99%), and (4) higher inter-school diversity in asset signal directions (Jaccard distance 0.853 vs. 0.778). These properties are necessary preconditions for the downstream Bayesian aggregation stage, which requires diverse, school-pure, and structurally parseable inputs.

Scope of current evidence. It is important to state clearly what these results do and do not establish. The evaluation demonstrates *school-style vocabulary consistency*, *output parsability*, *reduced cross-school contamination*, and *signal diversity*—all measured via keyword-based proxies on held-out synthetic events. The results do not yet establish that the fine-tuned adapters produce *better causal reasoning*, *better-calibrated probabilities*, or *superior trading performance* relative to the base model or any external benchmark. Those claims require Stages 3 and 4 of the pipeline, specifically the calibration evaluation against resolved Polymarket/Metaculus questions

and the out-of-sample backtests described in Chapter 5. The current results should therefore be interpreted as *input-quality validation* for Stage 3, not as evidence of end-to-end system performance.

Chapter 5

Future Plan and Schedule

Table 5.1 summarises the current completion status of each system component. Two components are complete: the episode database (Section 4.1) and the school-specific LoRA adapters (Section 4.2). Four components remain to be built and evaluated: the retrieval engine (Stage 1), the Bayesian aggregation and calibration layer (Stage 3), the asset allocation signal generator (Stage 4), and the end-to-end integration with ablation experiments. No end-to-end backtesting results are available at this stage.

Table 5.1: Component completion status.

Component	**Stage**	**Status**
Episode database (141 episodes, 35 FRED indicators, 22 instruments)	Stage 1 input	**Complete**
School-specific LoRA adapters (Keynesian, Austrian, Monetarist)	Stage 2	**Complete**
Dual-index retrieval engine (FAISS + Mahalanobis)	Stage 1	Planned
Bayesian aggregation and Platt scaling calibration	Stage 3	Planned
Fractional Kelly allocation and backtesting	Stage 4	Planned
End-to-end integration and ablation experiments	Full pipeline	Planned

Note that Stage 2 adapter training was completed prior to Stage 1 pipeline construction because the two components are independent: the LoRA adapters require only the episode narratives as seed content for trajectory generation, whereas the retrieval engine additionally requires the embedding index and Mahalanobis distance infrastructure implemented in Phase 1 below.

5.1 Stage 2: School-Specific LoRA Adapters (Completed)

Training of the three school-specific LoRA adapters (Keynesian, Austrian, Monetarist) on Qwen2.5-7B-Instruct is complete. Full details and evaluation results are reported in Chapter 4. No further work is required for this stage.

5.2 Stage 1: Retrieval-Augmented Analogy Engine

Embedding and indexing. We will encode all 141 episode narratives and their associated macro context vectors into a dual-index retrieval store. For the semantic channel, we will embed each episode's `embedding_text` field using a sentence transformer (e.g., `all-MiniLM-L6-v2` or `BGE-large-en`) and store the vectors in a FAISS index with inner-product search. For the structural channel, we will normalize the 35-dimensional FRED macro context vectors across all episodes, compute the pooled covariance matrix Σ, and implement Mahalanobis distance retrieval via $d_M(\mathbf{s}, \mathbf{s}_i) = \sqrt{(\mathbf{s} - \mathbf{s}_i)^\top \Sigma^{-1} (\mathbf{s} - \mathbf{s}_i)}$. Missing values in the macro context (primarily for pre-1990 episodes where certain FRED series do not exist) will be handled via mean imputation within each decade.

Retrieval fusion and evaluation. The semantic and structural scores will be combined via the weighted score in Eq. (3.1), with α tuned on a held-out validation set of 20 episodes. We will evaluate retrieval quality through two protocols: (i) a *category-match* metric measuring whether the top-3 retrieved episodes share at least one category tag with the query episode (e.g., a banking crisis retrieves other banking crises), and (ii) a *downstream utility* metric measuring whether including the retrieved analogies improves the school-specific causal chain quality compared to no-retrieval

baselines, scored by human or GPT-4o evaluation on a 1–5 rubric for causal coherence, mechanism specificity, and asset prediction accuracy.

Estimated timeline. 1–2 weeks. The primary engineering effort is building the dual-index retrieval pipeline and tuning α; no model training is required.

5.3 Stage 3: Bayesian Aggregation and Calibration

Calibration data collection. We will collect resolved prediction market questions from Polymarket (via public API and Kaggle dumps) and Metaculus (via the Python SDK at `github.com/Metaculus/forecasting-tools`), filtering for macroeconomic and geopolitical topics (monetary policy, inflation, GDP, elections, conflicts). We target a minimum of 500 resolved question–resolution pairs spanning 2023–2025 for calibration training and 200 held-out pairs for evaluation.

School-specific Brier Score tracking. For each resolved question, we will run all three LoRA adapters to generate predicted probabilities, then compute per-school Brier Scores partitioned by macro regime. Regime labels will be assigned by fitting a 3-state Gaussian HMM (using `hmmlearn`) on monthly S&P 500 returns and realized volatility, producing states interpretable as *expansion*, *contraction*, and *crisis*. The regime-conditional Brier Scores will populate the dynamic weighting function in Eq. (3.3).

Calibration layer training. We will implement and compare three calibration methods: Platt scaling (logistic, 2 parameters), isotonic regression (non-parametric), and temperature scaling [Guo et al., 2017] (1 parameter). Each will be trained on the 500-question training set and evaluated on the 200-question held-out set using Brier Score, Expected Calibration Error (ECE, 10 bins), and reliability diagrams. We will

report whether the two-stage process (regime-conditional weighting → Platt scaling) outperforms single-stage calibration applied directly to equal-weighted aggregates.

Estimated timeline. 2–3 weeks. The bottleneck is running all three adapters across 700+ questions and iterating on the calibration pipeline.

5.4 Stage 4: Asset Allocation Signal Generation and Backtesting

Signal construction. We will implement the fractional Kelly allocation from Eq. (3.6) with $f = 0.25$ and $w_{\max} = 0.10$ per asset. The system will produce signals for 8 liquid instruments: S&P 500 futures (ES), 10-year Treasury futures (ZN), gold futures (GC), WTI crude oil futures (CL), DXY index, EUR/USD, USD/JPY, and the iShares MSCI Emerging Markets ETF (EEM). Transaction costs will be modeled at 5 basis points per round trip for futures and 10 basis points for ETFs.

Backtesting protocol. Following the evaluation critique of Li et al. [2026] and walk-forward validation standards [López de Prado, 2018], we will adopt a strictly out-of-sample evaluation design:

- **In-sample period** (2015–2022): Used for calibration layer training, regime model fitting, and retrieval parameter tuning (α, f, λ). No signal-quality metrics will be reported on this period.
- **Out-of-sample period** (2023–2025): All reported Sharpe ratios, drawdowns, and directional accuracy are computed on this period only, using a rolling 6-month calibration window that expands forward.
- **Memorization defense**: Following Lopez-Lira et al. [2025], we will include a pre/post knowledge-cutoff performance comparison and apply counterfactual

perturbation tests (randomly shifting macro input values by ±10% and verifying that model predictions change proportionally rather than remaining anchored to memorized outputs).

- **Fine-tuning isolation**: LoRA adapter training relies on synthetic trajectories generated from abstract templates with randomized macro states, not on the historical episode records themselves. No instance-level information about specific 2023–2025 events enters the fine-tuning corpora, ensuring that the out-of-sample evaluation window is unseen at the adapter training stage.

Baselines. We will compare Macro-MOSAIC signals against five baselines: (i) buy-and-hold S&P 500, (ii) a random walk forecast (no-change assumption), (iii) a Bayesian VAR model (BVAR with Minnesota prior) on the same 8 instruments, (iv) GPT-4o with zero-shot CoT prompting given the same event inputs, and (v) the single best-performing LoRA adapter (no multi-school aggregation). Pairwise forecast comparisons will use Diebold-Mariano tests with Newey-West standard errors.

Metrics. Annualized Sharpe ratio, maximum drawdown, Sortino ratio, annualized return, directional accuracy (fraction of trades with correct sign), and CAPM alpha. Statistical significance of Sharpe differences will be assessed via the Ledoit and Wolf [2008] test.

Estimated timeline. 2–3 weeks. The primary effort is building the event-driven backtesting harness and ensuring no look-ahead bias in the signal generation pipeline.

5.5 End-to-End Integration and Ablation Experiments

Pipeline integration. We will assemble the four stages into a single callable pipeline: given a natural-language event description and a date, the system will automatically (1) fetch the current macro state from FRED, (2) retrieve top-3 historical analogies, (3) run all three LoRA adapters with the retrieved context to produce causal chains, (4) aggregate and calibrate probabilities, and (5) output cross-asset allocation signals. The pipeline will be implemented as a Python package with a CLI interface for reproducibility.

Ablation design. We will conduct systematic ablations to quantify each component's marginal contribution:

Table 5.2: Planned ablation experiments. Each row removes one component from the full system.

Ablation	What is removed	What it tests
−RAG	No historical analogy retrieval	Value of macro episode memory
−Multi-school	Single adapter (best-performing)	Value of school diversity
−Bayesian	Equal-weight aggregation	Value of regime-conditional weighting
−Calibration	Raw probabilities (no Platt)	Value of calibration layer
−Causal schema	Free-form CoT (no structured chain)	Value of structured transmission logic

All ablations will be evaluated on the same out-of-sample period (2023–2025) across all three evaluation dimensions: reasoning quality (EconCausal, EconLogicQA), calibration (Brier Score, ECE), and signal quality (Sharpe, drawdown). We will report 95% bootstrap confidence intervals on all metrics.

Reasoning evaluation. In parallel with the backtesting, we will evaluate causal chain quality on the EconCausal benchmark [Lee et al., 2025] (10,490 causal triplets) and EconLogicQA [Quan and Liu, 2024] (sequential economic reasoning). This evalu-

ation isolates the reasoning contribution from the financial performance and provides a direct comparison against GPT-4o and Claude Sonnet.

5.6 Timeline Summary

Table 5.3: Projected timeline for remaining work.

Phase	**Task**	**Duration**
Completed	Episode database construction (141 episodes)	—
Completed	School-specific LoRA adapter training (×3)	—
Phase 1	RAG engine: embedding, indexing, retrieval fusion	Weeks 1–2
Phase 2	Calibration data collection & Bayesian aggregation	Weeks 2–4
Phase 3	Signal generation & walk-forward backtesting	Weeks 4–6
Phase 4	End-to-end integration & ablation experiments	Weeks 6–8
Phase 5	Reasoning benchmarks (EconCausal, EconLogicQA)	Weeks 7–8
Phase 6	Paper writing & supplementary materials	Weeks 8–10

5.7 Risk, Governance, and Auditability

Deploying an LLM-based system in a financial decision context raises several risks that we address explicitly here.

Hallucination and factual grounding. LLMs can generate plausible but factually incorrect causal chains. Macro-MOSAIC mitigates this through two structural mechanisms: (i) retrieved historical episodes provide concrete anchor points—specific dates, FRED values, and measured asset reactions—that constrain generation, and (ii) the structured chain schema requires each step to name a specific, verifiable transmission mechanism (e.g., credit channel, exchange rate channel) rather than vague narrative. Causal claims are therefore traceable to specific evidence in the episode database. Residual hallucination risk will be quantified in the EconCausal evaluation

(Section 5.4), which compares generated causal steps against a ground-truth causal triplet database.

Data leakage and reproducibility. As discussed in Section 5.3, LoRA adapters are trained on synthetic trajectories generated from abstract templates with randomised macro snapshots; no historical episode records from the out-of-sample period (2023–2025) enter the fine-tuning corpora. The episode database construction pipeline will be fully documented and released, including data sources (Bloomberg, Yahoo Finance, FRED), extraction scripts, and the LLM prompts used to generate narrative summaries and school interpretations, enabling independent replication.

Backtest overfitting. Walk-forward evaluation with a rolling 6-month calibration window and strictly out-of-sample reporting (Section 5.3) limits look-ahead bias. In addition, the −RAG and −Multi-school ablations (Section 5.4) test whether performance gains are attributable to the structural components or to incidental pattern matching. We will apply the Ledoit-Wolf Sharpe ratio test and report bootstrap confidence intervals to guard against inflated significance claims.

Human oversight. Macro-MOSAIC is designed as a decision-support system, not an autonomous trading agent. Each output is a structured record containing the retrieved analogies, explicit causal chains, scenario probabilities, and per-asset conviction scores, all of which a human portfolio manager can inspect and override. The explicit school labels and transmission mechanism fields make the system's reasoning auditable in a way that black-box LLM outputs are not. In any live deployment, final allocation decisions would remain under human review, with the system's role limited to structured signal generation.

Traceability. The causal chain schema (event → mechanism → intermediate effects → asset impacts) ensures that every allocation signal can be traced back through the

probability aggregation to the school-specific chains, and from those chains to the retrieved historical episodes and the specific FRED values and price reactions that motivate the analogy. This traceability is an explicit design objective of Macro-MOSAIC and distinguishes it from systems that produce opaque end-to-end predictions.

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